

B.E. (Mechanical), III Year V SEMESTER (2024-25)
TEACHING AND EXAMINATION SCHEME

S.N	Course Category	Course Code	Course Title	Credits			Contact Hours/week	Total Credits	Exam Hours (ETE)	Marks				
				L	T	P				Theory		Practical		Total
										IA	ETE	IA	ETE	
1	PCC	5ME41A(ME)	Refrigeration & Air Conditioning	2	1	-	3	3	3	30	70	-	100	
2	PCC	5ME42A(ME)	Manufacturing technology	3	-	-	3	3	3	30	70	-	100	
3	PCC	5ME43A(ME)	Dynamics of Machines	3	1	-	4	4	3	30	70	-	100	
4	PCC	5ME44A(ME)	Operations Research	3	-	-	3	3	3	30	70	-	100	
5	PEC	5ME51A(ME) 5ME52A(ME) 5ME53A(ME) 5ME54A(ME)	PROFESSIONAL ELECTIVE COURSE- I (Students to select any one course offered by the department)	3	-	-	3	3	3	30	70	-	100	
6	OEC		OPEN ELECTIVE COURSE – II (Students to select course offered by other department)	3	-		3	3		--	100		100	
	Total (Theory)			17	2	-	19	19	-	150	450	-	600	
7	PCC	5ME41B(ME)	Thermal Engineering Lab. II	-	-	1.5	3	1.5	3	-		50	50	100
8	PCC	5ME42B(ME)	Theory of Machines Lab	-	-	1	2	1	3	-		50	50	100
9	PCC	5ME43B(ME)	Production Engineering Lab. III	-	-	1.5	3	1.5	3	-		50	50	100
10	PCC	5ME44B(ME)	Computer Aided Design Lab.	-	-	1	2	1	3	-		50	50	100
	Total (Practical)			-	-	5	10	5	-	-	-	200	200	400
11	PSI	5ME71B(ME)	Practical Training I (At the end of IV semester)					1				100	100	
	Grand Total (Theory+Practical+training)			17	2	5	29	25	-	150	450	500	1100	
12	MNC	Co-curricular Activities (NCC/NSS/CLUB/YOGA/SCOUT etc.)					-	-	-	-	-	-	-	

L:Lecture, T:Tutorial, P:Practical, ETE: End Term Examination, IA: Internal assessment
Evaluation of co-curricular activities will be done for V and VI semester at the end of VI semester.

5ME41A(ME): REFRIGERATION AND AIRCONDITIONING

2L, 1T

ETE: 70 marks, IA: 30 marks

COURSE OUTCOMES:

CO1: Understand various air-refrigeration cycles for aircraft refrigeration.

CO2: Analyze vapor compression system at various operating conditions.

CO3: Understand the CFC, HCFC, HFC and HC types of refrigerants and their properties.

CO4: Demonstrate the concept of vapor absorption refrigeration system and Electrolux refrigerator.

CO5: Understand the psychrometry of moist air and its measurement.

CO6: Psychrometry of summer and winter air-conditioning processes and its use in comfort air-conditioning.

SYLLABUS:

Refrigeration principle: Second law interpretation of heat engine, heat pump and refrigerator; reversed Carnot cycle, coefficient of performance, limitations of reversed Carnot cycle, unit of refrigeration. Air Refrigeration System: Bell-Coleman cycle; air cycle systems for aircraft refrigeration; simple system, boot strap system, regenerative system and reduced ambient system.

Vapour Compression System: Theoretical vapour compression systems, cycle analysis using p-h diagram; Effect of operating conditions, suction vapour superheat and liquid sub cooling. Refrigerants: Desirable properties, designation and comparative study of refrigerants; substitutes for CFC refrigerants.

Vapour absorption system: Simple vapour absorption system, common refrigerant-absorbent combinations; Electrolux refrigerator.

Refrigeration Equipments: Types of compressors, condensers, evaporators and expansion devices and their working; cooling tower; control and protection devices.

Psychrometry: Thermodynamics properties of moist air, perfect gas relationship for approximate calculation; psychrometric properties, DBT, DPT, humidity ratio, degree of saturation, relative humidity, enthalpy, WBT, Adiabatic saturation process, psychrometric chart.

Psychrometry of air conditioning processes: Cooling and dehumidification, sensible cooling with dry coils, direct expansion wet coils, apparatus dew point, by pass factor; heating and humidification, heating coils; Air Washers; chemical dehumidification; water and steam injection; Summer and Winter air conditioning. Introduction to comfort air conditioning, effective temperature and comfort charts.

5ME42A(ME): MANUFACTURING TECHNOLOGY

3L

ETE: 70 marks, IA: 30 marks

COURSE OUTCOMES:

CO1: Understand plastic deformation criteria and power requirement in metal forming.

CO2: Understand various processes of plastic deformation in metal forming.

CO3: Classify modern machining methods and explain controlling parameter on MRR.

CO4: Illustrate typical manufacturing processes to produce components e.g. shafts, gears and threads.

CO5: Understand the structure of various computer-controlled technology used in manufacturing technology.

CO6: Develop the ability to do NC and CNC programming.

SYLLABUS:

UNIT – 1

Metal Working: Metal behaviour in metal forming – Principal stresses and strains, Plastic deformation and Yielding criteria; Hot and Cold working; Friction and Lubrication in metal forming.

UNIT – 2

Bulk Deformation Processes: Forging – Open and Closed die, Impression, Drop, Upset and Press forging; Rolling – Flat and Shape rolling; Extrusion – Forward, Backward and Impact extrusion; Wire, Rod and Tube drawing. Sheet Metal Working: Cutting – Shearing, Blanking, Punching; Drawing; Bending; Spinning; High energy rate forming.

UNIT - 3

Modern Machining Methods: Classification; Abrasive and Water jet; Ultrasonic; Electrochemical; Chemical milling; Electric discharge; Plasma arc machining.

UNIT - 4

Production of Machine Components: Processes of producing Shafts; Threads – Chasing, Rolling, Dies and Taps, Milling, grinding; Gears – Casting, Stamping, Rolling, Milling, Shaping, Hobbing, Shaving and Grinding.

UNIT – 5

Rapid Prototyping: Techniques; Applications, Robotics and Mechatronics, Flexible Manufacturing Systems: NC part programming – Manual and Computer assisted (APT); Concept of flexibility; Computer integrated manufacturing systems.

5ME43A(ME): DYNAMICS OF MACHINES

3L, 1T

ETE: 70 marks, IA: 30 marks

COURSE OUTCOMES:

CO1: Understanding dynamic force analysis of mechanisms and machines.

CO2: Analyze the position, velocity and acceleration of links of different slider crank mechanisms.

CO3: Understand various types of gears, gear profiles and gear trains and their analysis.

CO4: Analyze the balancing problems in rotating and reciprocating machinery.

CO5: Analyze free, forced and damped vibration characteristics of single degree of freedom systems.

SYLLABUS:

UNIT 1

Inertia Torque: Inertia forces of a floating link and a link rotating about a fixed centre, Four-bar Mechanism, Combined static and inertia force analysis.

UNIT 2

Dynamics of slider crank mechanism; Displacement, velocity and acceleration of slider, Velocity and acceleration of connecting rod, Crank effort, dynamically equivalent Systems.

UNIT 3

Gear: Definition and law of gearing, Velocity of sliding between two teeth in mesh, Involute and cycloidal profiles for gear teeth, Interference and its checking, helical, Spiral, bevel and worm gear

Gear Train: Simple, compound and epicyclic trains including Humpage's gear, Gear train Applications

UNIT 4

Balancing: Balancing of rotating masses in one and different planes, Balancing of reciprocating masses, Balancing of inline and V- engines, Balancing machines

UNIT 5

Vibrations: Free and forced vibrations of single degree of freedom systems, effect of damping, vibration isolation, resonance, critical speed of shafts.

5ME44A(ME): OPERATIONS RESEARCH

3L

ETE: 70 marks, IA: 30 marks

COURSE OUTCOMES:

CO1: Identify and Apply operations research techniques in industrial optimization problem.

CO2: Analyze to formulate a real-world problem as a mathematical linear programming model.

CO3: Apply linear programming to solve transportation and assignment problems.

CO4: Apply game theory to evaluate best strategy in decision making.

CO5: Analyze the problems of Inventory control using various models.

CO6: Understand various models adopted for queuing theory problems.

SYLLABUS:

UNIT 1

Operation research: Characteristics; Scope; Methodologies; Modeling.

UNIT 2

Linear Programming: Problem formulation; Graphical and Simplex methods – Restricted or unrestricted variables, Equality or inequality constraints, Degeneracy, Alternative optima, Unbounded space and solution, Infeasibility; Duality – Primal-dual relationship; Sensitivity analysis – Changes in resource and contribution coefficients

UNIT 3

Transportation and Assignment: LP formulation; Transportation algorithm; Hungarian algorithm.

Game theory: Pure and mixed strategies; Graphical solution; Dominance; LP formulation

UNIT 4

Inventory Control: Types of inventories; Relevant costs; Static inventory decisions under risk– Payoff, Opportunity cost, Expected value of perfect information; Uncertainty – decision criteria; Dynamic inventory models under certainty – EOQ with finite or infinite rate of replenishment, with or without shortages, price breaks; Probabilistic inventory models safety stocks, re-order levels; Selective control of inventories.

UNIT 5

Queuing Theory: Elements; Measures of performance; (M/M/1):(FCFS/ ∞/∞) model.

Simulation: Monte Carlo technique; Applications to inventory and queuing situations. Elementary Concept of CPM and PERT

5ME51A(ME): INTERNAL COMBUSTION ENGINES

3L

ETE: 70 marks, IA: 30 marks

COURSE OUTCOMES:

CO1: Understand combustion process, detonation, carburetion and fuel injection in S.I. engines.

CO2: Understand combustion process and knocking phenomenon in C.I. engines.

CO3: Measure the performance of I.C. Engines by various performance parameters and heat balance sheet.

CO4: Understand the phenomenon of Supercharging, its methods and types in I.C. Engines.

CO5: Illustrate various norms related to air pollution from vehicular emission and methods of its control.

SYLLABUS:

S.I. Engines: Deviation from the ideal cycle, combustion losses, flame development and propagation, effects of operating variable on detonation; Detonation rating of engine fuels, HUCR Dope, preignition combustion chambers, Mechanics of carburetion, MPFI systems, mixture requirement and combustion.

C.I. Engines: Stages of combustion, Diesel knock, effect of operating variables, knock rating of C.I. engine fuels, Cetane number; Effect of additives, combustion chambers. Requirements of fuel injection system and its mechanism, Fuel filters.

Lubrication and Cooling: Forced, splash, by-pass and direct, Blow by and crankcase dilution.

Air and water cooling, Thermostatic control.

Performance and Testing: High speed indicators, various efficiencies, Morse Test and Willan's line; torque and mean effective pressure, performance and heat balance sheet, Measurement of volumetric efficiency, Effect of atmospheric condition on performance of I.C. Engines; Supercharging; high altitude problem of I.C. Engines, methods and types of supercharging; suitability of S.I. and C.I. Engines;

Air Pollution: Vehicular emission, its norms, Causes and control of air pollution from diesel and petrol engines, catalytic converter.

Introduction to dual fuel and multi-fuel engines; free-piston engine and rotary combustion engine.

5ME52A(ME): COMPOSITE MATERIALS

3L

ETE: 70 marks, IA: 30 marks

COURSE OUTCOMES:

- CO1:** Understand fundamentals, types and application of composites.
- CO2:** Explain the characteristics and manufacturing of Metal matrix composites.
- CO3:** Explain the characteristics and manufacturing of Ceramic matrix composites.
- CO4:** Understand design of composites for structural and functional applications.
- CO5:** Explain the characteristics and manufacturing of functionally gradient composites.

SYLLABUS:

INTRODUCTION TO COMPOSITES: Fundamentals of composites - need for composites – Enhancement of properties - classification of composites – Matrix-Polymer matrix composites (PMC), Metal matrix composites (MMC), Ceramic matrix composites (CMC) – Reinforcement – Particle reinforced composites, Fibre reinforced composites. Applications of various types of composites. Structure and mechanical properties; characterization of fibres and weibull analysis;

METAL MATRIX COMPOSITES: Characteristics of MMC, Various types of Metal matrix composites Alloy vs. MMC, Advantages of MMC, Limitations of MMC, Metal Matrix, Reinforcements – particles – fibres. Effect of reinforcement - Volume fraction – Rule of mixtures. Processing of MMC – Powder metallurgy process - diffusion bonding – stir casting – squeeze casting.

CERAMIC MATRIX COMPOSITES: Ceramic matrix - Various types of Ceramic Matrix composites- oxide ceramics – non oxide ceramics – aluminium oxide – silicon nitride – reinforcements – particles- fibres- whiskers. Sintering - Hot pressing – Cold isostatic pressing (CIPing) – Hot isostatic pressing (HIPing).

Overview of creep, fatigue and fracture of composites, role of matrix reinforcement interfaces and interface engineering, design of composites for structural and functional applications. Laminated composites and multilayered coatings, Theories of composite strengthening, Fracture and Toughening Mechanisms, Functionally gradient composites; strengthening mechanisms in FGMs, continuum mechanics (macro and micro)

5ME53A(ME): FOUNDRY AND WELDING ENGINEERING

3L

ETE: 70 marks, IA: 30 marks

COURSE OUTCOMES:

CO1: Understand foundry practices, basic terminology and equipment.

CO2: Classify, compare and contrast among various casting processes.

CO3: Classify, compare and contrast among various welding processes.

CO4: Understand various welding defects and testing of welded joints.

SYLLABUS:

UNIT 1

Foundry: Mould making; Pattern – types, allowances; Moulding sand – ingredients, additives, properties, methods of testing; Core materials. Casting Design: Metal solidification, cooling, shrinkage, directional solidification; Chaplet and Chill, Gate, Runner and Riser; Product design considerations.

UNIT 2

Expendable Mould Casting methods: Sand casting – Green, Dry, Loam sand moulding, Carbon dioxide moulding, Pit and Floor moulding; Shell moulding; Investment casting; Plaster and Ceramic mould casting. Permanent Mould Casting Methods: Die casting; Centrifugal casting; Continuous casting.

UNIT 3

Foundry Practices: Equipment, machines and mechanization; Melting furnaces – constructional features and operation of Cupola, Electric furnace; Fettling, cleaning and inspection; Casting defects and remedies; Sand reclamation.

UNIT 4

Welding: Metal coalescence; Types of joints, welds and welding positions; Physics and metallurgy of welding; Weldability. Brazing and Soldering: Methods and materials. Fusion Welding: (i) Arc Welding: Principle; Electrodes, welding machines, tools and power source – current, voltage and polarity; Constant current and Constant potential machines; Shielded metal arc welding; Gas metal arc (MIG); Flux-cored arc; Electrode gas; Submerged arc. Non-consumable electrode arc welding processes – Gas tungsten arc (TIG); Plasma arc; Carbon arc and Stud welding. (ii) Resistance Welding: Spot, Seam, Projection, Flash butt, Upset, Percussion, High-frequency resistance welding; Thermal spraying. (iii) Oxyacetylene Gas Welding: Heat generation and flame; Torch and other equipment; Gas cutting; Oxyhydrogen welding. (iv) Electron beam; Laser beam; Electroslag; Thermit welding.

UNIT 5

Introduction to Solid-State Welding: Forge; Roll; Hot pressure; Diffusion; Explosive; Friction; Ultrasonic welding. Weld Quality: Residual stresses and distortion; Heat affected zone; Welding defects and remedies; Inspection and Non-destructive testing – Dye penetrates; Magnetic particles; Ultrasonic; Radiographic

5ME54A(ME): FUNDAMENTALS OF HVAC SYSTEMS

3L

ETE: 70 marks, IA: 30 marks

COURSE OUTCOMES:

CO1: Understand and estimate heat gain through a building structure.

CO2: Determine the internal and system heat loads with separate sensible and latent heat.

CO3: Make use of applied psychrometry to estimate dehumidified air quantity.

CO4: Analyze the various HVAC systems components and selection.

CO5: Understand duct design for proper room air distribution.

Heat Transfer Through Building Structures: Building Survey, Periodic heat transfer through walls and roofs. Empirical methods to calculate heat transfer through walls and roofs; Heat gain through glass, calculation of solar heat gain through ordinary glass tables, shading devices, effect of shading devices. Equivalent temperature difference method, Thermal resistance of various building materials, Infiltration, stack effect, wind effect.

Heating and cooling load Calculations: Outside and Inside design conditions, Cooling load estimate, heating load estimate, heat losses through structure, heat losses due to infiltration, ventilation, Effects of solar radiation, internal heat gains and system heat gains on heating load calculations,

Applied psychometrics: Simple air-conditioning system, summer air-conditioning, psychometric calculations for cooling, ventilation air, bypass factor, Grand sensible heat factor, effective sensible heat factor, winter air-conditioning.

HVAC System components and selection: HVAC system component and selection criteria including cooling coils, room air distribution, AHU, fans and air circulation, humidifying and dehumidifying processes, piping and ducting design, air quality standards, Air Filters, Duct Design and installation, Testing and Balancing Air Systems. Chilled Water Systems, Primary and secondary water loops, constant and variable flow; Cooling Towers.

5ME41B(ME):THERMAL ENGINEERING LAB. II

3P

ETE: 50 marks, IA: 50 marks

COURSE OUTCOMES:

- CO1:** Understand and analyze the performance of vapour compression refrigeration systems with its accessories.
- CO2:** Understand Vapour absorption refrigeration system and Electrolux refrigerator.
- CO3:** Analyze the performance of diesel engine with the help of heat balance sheet.
- CO4:** Understand air-conditioning systems of a room, passenger car and train compartment.
- CO5:** Demonstrate the use of Morse test to calculate Indicated power of a multicylinder petrol engine.

LIST OF EXPERIMENTS:

1. To study vapour compression refrigeration systems (domestic refrigerator, water cooler) and its components (Compressor, condenser, evaporator, throttling device, automatic and thermostatic expansion valves, capillary tube, thermostatic switch)
2. To study vapour compression refrigeration accessories (liquid receiver, sight glass, drier, solenoid valve, LP cut-off switch, HP cut-off switch, Rotameter)
3. To study window type one-ton air conditioner and its comparison with split air-conditioner
4. To study air-conditioning system of a modern passenger car and a train compartment.
5. To study 600 kg ice plant and its accessories.
6. To study vapour absorption cycle refrigerator (Electrolux refrigerator).
7. To conduct a load test on water calorimeter and calculate ideal and actual COP of the system and Horse power required to drive the compressor.
8. To conduct a test on vortex tube and calculate adiabatic efficiency and COP.
9. To conduct a constant speed load test on Kirloskar diesel engine. Measure fuel and air consumption and prepare a heat balance sheet.
10. To perform Morse test on petrol engine and draw the performance curves.

5ME42B(ME): THEORY OF MACHINES LAB

2P

ETE: 50 marks, IA: 50 marks

COURSE OUTCOMES:

- CO1:** Build the concept of Gyroscope motion.
- CO2:** Understand the use of wind tunnel and 6" mirror type Schliern system.
- CO3:** Determine practically the frequency torsion vibration.
- CO4:** Determine practically the critical speed of whirling shaft.
- CO5:** Understand the purpose of governors and their types
- CO6:** Classify and demonstrate the use of various types of dynamometers.

LIST OF EXPERIMENTS:

1. To study Gyroscope and discuss the relation between precession angular speed.
2. To study wind tunnel including its traverse mechanism.
3. To study and find natural frequency of torsion vibration. Also find the frequency for free damping.
4. To study and determine the critical speed of whirling shaft.
5. To study 6" mirror type Schliern system.
6. To study various types of governors.
7. To study various types of dynamometer (Rope brake, Froude hydraulic, swinging field electric and Eddy current dynamometer).

5ME43B(ME):PRODUCTION ENGINEERING LAB III

3P

ETE: 50 marks, IA: 50 marks

COURSE OUTCOMES:

CO1: Develop the knowledge of application and analysis of orthogonal metal cutting.

CO2: Analyze and evaluate the effect of metal cutting parameters on the process.

CO3: Identify the influence of different factors on machinability.

CO4: Study the machine tools and precision measurement.

CO5: Explain practically alignment operations on lathe, milling and drilling machines to develop capability of machine.

LIST OF EXPERIMENTS:

METAL CUTTING LAB

1. To study and calibrate a 2D lathe dynamometer.
2. To plot tangential and feed force as function of cutting speed and feed in turning, also calculate and plot cutting power as a function of feed rate.
3. To study the effect of rake angle and chip thickness ratio on shear angle in orthogonal machining.
4. To perform square thread cutting and prepare its process chart.
5. To perform alignment test on center lathe.

MACHINE TOOL LAB

6. Explain term machinability and conduct cuttability test on a drilling machine for comparing machinability of different metals.
7. To study the tool makers microscope and checking the alignments of screw thread and cutting tool (five elements of workpiece are checked, outer dia, root dia, pitch, depth, thread angle) (measuring angles end cutting edge angle, side cutting edge angle, lip angle, top back rake angle, clearance angle).
8. To calibrate the Pneumatic comparator and measure angles and taper of given test materials.
9. To perform alignment test on a drilling machine, milling machine and surface grinder.

5ME44B(ME): COMPUTER AIDED DESIGN LAB

2P

ETE: 50 marks, IA: 50 marks

COURSE OUTCOMES:

CO 1: Demonstrate basic concepts of the AutoCAD software, construction (drawing) and dimensioning technique.

CO 2: Explain to manipulate drawings through editing and plotting techniques

CO 3: Understand to do velocity and acceleration analysis.

CO 4: Develop 2D Orthographic Projections.

CO 5: Develop 3D solid modeling.

List of Experiments:

1. Overview of AutoCAD.
2. Drawing Commands, Edit Commands, View Commands, Drawing Settings, dimensioning, setting dimensioning variables, saving and plotting.
3. Involute profile, velocity analysis and acceleration analysis in AutoCAD.
4. 2D Orthographic Projections in AutoCAD.
5. Assembly making in AutoCAD.
6. 3D solid modeling in AutoCAD.